JavaScript Assignment 03

Arrays, Loops, Flags & Problem Solving

Course:	Web & Mobile App Development
Subject:	JavaScript
Trainer:	Sir Jaffar Aman
Assignment Number:	03
Reference Book:	A Smarter Way to Learn JavaScript — Mark Myers
Chapters:	15–20
Total Marks:	100
Duration:	7 Days
Submission:	GitHub Repository

Assignment Objective

This assignment is designed to strengthen your JavaScript problem-solving skills through arrays, loops, nested loops, conditions, and flag-based searching.

After completing this assignment, you should be able to:

- Generate number sequences using loops.
- Work with forward and reverse loops.
- Access and process array elements.
- Filter array values using loops and conditions.
- Search for values using flags and `break`.
- Solve logical problems using nested loops.
- Create patterns and small real-world programs.

General Instructions

1. Complete all 20 questions.
2. Use JavaScript `for` loops wherever iteration is required.
3. Use arrays and conditional statements where appropriate.
4. Use `prompt()` for questions requiring user input.
5. Display output using `console.log()`, `alert()`, or `document.write()`.
6. Do not use `map()`, `filter()`, `reduce()`, `forEach()`, `sort()`, `reverse()`, `includes()`, or `indexOf()` for solving these problems.
7. Do not use functions, objects, or DOM manipulation.
8. Use meaningful variable names and proper indentation.
9. Write your own logic instead of copying solutions.
10. Push your completed assignment to GitHub.

Section A — Loop Fundamentals (20 Marks)

Question 01 — Counting Generator (4 Marks)

Write a JavaScript program that prints numbers from 1 to 100 using a `for` loop.

Expected Output:

```
1 2 3 4 5 ... 100
```

Question 02 — Reverse Counting (4 Marks)

Write a program that prints numbers from 50 down to 1.

Expected Output:

```
50 49 48 47 ... 3 2 1
```

Question 03 — Even & Odd Number Generator (4 Marks)

Write a program that prints all even and odd numbers separately between 1 and 50.

Expected Output:

```
Even: 2 4 6 8 ... 50
Odd: 1 3 5 7 ... 49
```

Question 04 — Multiplication Table (4 Marks)

Ask the user to enter a number and generate its multiplication table from 1 to 10.

Sample Input:

```
Enter a number: 7
```

Expected Output:

```
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
...
7 x 10 = 70
```

Question 05 — Sum of Natural Numbers (4 Marks)

Ask the user to enter a positive integer `N`.

Calculate the sum of all numbers from 1 to N using a loop.

Sample Input:

```
N = 10
```

Expected Output:

```
Sum = 55
```

Section B — Loops with Arrays (25 Marks)

Question 06 — Display Array Elements (5 Marks)

Create an array containing eight student names.

Display each name with its position number using a loop.

Expected Output:

```
1. Ali
2. Ahmed
3. Sara
4. Hamza
...
8. Ayesha
```

Question 07 — Find Largest & Smallest (5 Marks)

Given the following array:

```
var numbers = [45, 12, 89, 3, 67, 22, 91, 10];
```

Find the largest and smallest numbers using loops.

Expected Output:

```
Largest: 91
Smallest: 3
```

Do not use `Math.max()`, `Math.min()`, or `sort()`.

Question 08 — Reverse an Array (5 Marks)

Given:

```
var fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"];
```

Display all elements in reverse order using a loop.

Expected Output:

```
Grapes
Orange
Mango
Banana
Apple
```

Do not use the `reverse()` method.

Question 09 — Filter Passing Students (5 Marks)

Given:

```
var marks = [85, 42, 67, 30, 90, 55, 25, 78];
```

Create a new array containing only marks greater than or equal to 50.

Expected Output:

```
Passing Marks: [85, 67, 90, 55, 78]
Total Passed: 5
```

Use a loop, conditions, and `push()`.

Do not use the `filter()` method.

Question 10 — Sum & Average of Array (5 Marks)

Given:

```
var prices = [1200, 850, 430, 1600, 990];
```

Calculate the total and average price.

Expected Output:

```
Total: 5070
Average: 1014
```

Section C — Flags, Searching & Filtering (20 Marks)

Question 11 — Student Name Search (5 Marks)

Given:

```
var students = ["Ali", "Ahmed", "Sara", "Hamza", "Ayesha"];
```

Ask the user to enter a student name.

Search the array using a `for` loop and a Boolean flag.

Example:

```
Enter Name: Sara
Student Found!
```

If the name does not exist:

```
Student Not Found!
```

Use `break` when the name is found. Match names exactly.

Question 12 — Find a Number (5 Marks)

Given:

```
var numbers = [10, 25, 40, 55, 70, 85, 100];
```

Ask the user to enter a number.

Search the array and display its index if found.

Example:

```
Enter Number: 55
Number found at index 3.
```

Otherwise:

```
Number not found.
```

Question 13 — Filter Numbers (5 Marks)

Given:

```
var numbers = [12, -5, 0, 18, -9, 25, 0, -3, 40];
```

Use loops and conditions to create three separate arrays:

- Positive numbers
- Negative numbers
- Zeros

Expected Output:

```
Positive: [12, 18, 25, 40]
Negative: [-5, -9, -3]
Zeros: [0, 0]
```

Question 14 — Remove Duplicate Numbers (5 Marks)

Given:

```
var numbers = [10, 20, 10, 30, 20, 40, 30, 50];
```

Create a new array containing only unique values.

Expected Output:

```
[10, 20, 30, 40, 50]
```

Use loops, conditions, and a flag to check whether a value already exists.

Do not use `Set`, `includes()`, or `indexOf()`.

Section D — Nested Loops & Patterns (20 Marks)

Question 15 — Star Triangle (5 Marks)

Use nested loops to print the following pattern:

```
*
**
***
****
*****
```

Question 16 — Reverse Star Triangle (5 Marks)

Use nested loops to print:

```
*****
****
***
**
*
```

Question 17 — Number Triangle (5 Marks)

Use nested loops to generate:

```
1
12
123
1234
12345
```

Question 18 — Multiplication Table Grid (5 Marks)

Use nested loops to generate multiplication tables from 2 to 5.

Each table should contain multiplication from 1 to 10.

Example:

```
Table of 2
2 x 1 = 2
...
2 x 10 = 20

Table of 3
3 x 1 = 3
...
3 x 10 = 30
```

Continue until the table of 5.

Section E — Real-World Logic Challenges (15 Marks)

Question 19 — Student Result Analyzer (7 Marks)

A teacher has the following student data:

```
var students = ["Ali", "Sara", "Ahmed", "Ayesha", "Hamza"];
var marks = [78, 92, 45, 88, 33];
```

Each student's name and marks are stored at the same index.

Requirements:

1. Display every student's name and marks.
2. Consider marks of 50 or above as PASS.
3. Count total passed students.
4. Count total failed students.
5. Find the highest-scoring student.
6. Calculate the class average.

Expected Output:

```
Ali - 78 - PASS
Sara - 92 - PASS
Ahmed - 45 - FAIL
Ayesha - 88 - PASS
Hamza - 33 - FAIL

Total Students: 5
Passed: 3
Failed: 2
Top Student: Sara (92)
Class Average: 67.2
```

Question 20 — Mini Shopping Cart (8 Marks)

Create a shopping cart using the following arrays:

```
var products = ["Keyboard", "Mouse", "Monitor", "USB", "Headphones"];
var prices = [2500, 1200, 22000, 800, 3500];
```

Requirements:

1. Display all products with their prices and product numbers (1–5).
2. Ask the user to select a product number.
3. Ask the user to enter the quantity.

4. Validate the product selection and quantity.
5. Calculate the total amount.
6. If the total is Rs. 5,000 or more, apply a 10% discount.
7. Display the final bill.

Example Input:

```
Product Number: 1
Quantity: 3
```

Expected Output:

```
Product: Keyboard
Unit Price: Rs. 2500
Quantity: 3
Subtotal: Rs. 7500
Discount: Rs. 750
Final Bill: Rs. 6750
```

Note: This program handles one selected product per execution. Multiple-product carts are not required.

Bonus Challenges (Optional — Not Included in 100 Marks)

Bonus 01 — Prime Number Checker

Ask the user to enter a positive integer greater than 1.

Determine whether it is prime using a loop and a flag.

Bonus 02 — Common Elements

Given:

```
var array1 = [10, 20, 30, 40, 50];
var array2 = [30, 40, 50, 60, 70];
```

Find common elements using nested loops.

Expected Output:

```
[30, 40, 50]
```

Bonus 03 — Frequency Counter

Given:

```
var numbers = [2, 3, 2, 5, 3, 2, 4];
```

Count how many times each unique number appears using nested loops.

Expected Output:

```
2 appears 3 times
3 appears 2 times
5 appears 1 time
4 appears 1 time
```

GitHub Submission Guidelines

Repository: `SMIT-Batch22-JS-Assignments`

Required Structure:

```
SMIT-Batch22-JS-Assignments/
|-- Assignment-01/
|-- Assignment-02/
`-- Assignment-03/
    |-- index.html
    |-- question-01.js
    |-- question-02.js
    |-- ...
    |-- question-20.js
    `-- README.md
```

Submit your GitHub repository link with all 20 completed questions.

Include your name, batch number, trainer name, and screenshots of at least five program outputs in the README.

Evaluation Criteria

Criteria	Marks
Section A — Loop Fundamentals	20
Section B — Loops with Arrays	25
Section C — Flags & Filtering	20
Section D — Nested Loops	20
Section E — Real-World Challenges	15
Total	**100**

Code readability, correct outputs, independent problem-solving, and GitHub submission will be considered while awarding marks.

Final Instructions

- Do not use advanced JavaScript methods to bypass loop logic.
- Test your programs with different inputs.
- Write clean and readable code.
- Understand every line you write.
- Be prepared for a short practical viva.

Best of Luck, Batch 22!

Trainer: Sir Jaffar Aman

Don't just write code. Learn to think like a programmer.